

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/307896924>

Effects of Velocity Loss During Resistance Training on Performance in Professional Soccer Players

Article in *International Journal of Sports Physiology and Performance* · September 2016

DOI: 10.1123/ijspp.2016-0170

CITATIONS

81

READS

6,307

4 authors, including:



[Fernando Pareja Blanco](#)

Universidad Pablo de Olavide

99 PUBLICATIONS 1,701 CITATIONS

[SEE PROFILE](#)



[Luis Suarez-Arrones](#)

Universidad Pablo de Olavide

122 PUBLICATIONS 1,965 CITATIONS

[SEE PROFILE](#)



[Juan José González-Badillo](#)

Universidad Pablo de Olavide

181 PUBLICATIONS 7,794 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



Training Interventions to Improve Soccer Players Performance [View project](#)



Velocity-based resistance training [View project](#)

Note. This article will be published in a forthcoming issue of the *International Journal of Sports Physiology and Performance*. The article appears here in its accepted, peer-reviewed form, as it was provided by the submitting author. It has not been copyedited, proofread, or formatted by the publisher.

Section: Original Investigation

Article Title: Effects of Velocity Loss During Resistance Training on Performance in Professional Soccer Players

Authors: Fernando Pareja-Blanco, Luis Sánchez-Medina, Luis Suárez-Arrones, and Juan José González-Badillo

Affiliations: Faculty of Sport, Pablo de Olavide University, Seville, Spain.

Journal: *International Journal of Sports Physiology and Performance*

Acceptance Date: August 1, 2016

©2016 Human Kinetics, Inc.

DOI: <http://dx.doi.org/10.1123/ijsp.2016-0170>

EFFECTS OF VELOCITY LOSS DURING RESISTANCE TRAINING ON PERFORMANCE IN PROFESSIONAL SOCCER PLAYERS

Fernando Pareja-Blanco
Luis Sánchez-Medina
Luis Suárez-Arrones
Juan José González-Badillo

Type of article: ORIGINAL INVESTIGATION

Contact author: Fernando Pareja-Blanco

Facultad del Deporte, Universidad Pablo de Olavide, Ctra. de Utrera, km 1, 41013 Seville,
SPAIN

Tel + 34 653 121 522, Fax: +34 954 348 659, email: fparbla@gmail.com

Preferred running-head: VELOCITY LOSS AS A RESISTANCE TRAINING
VARIABLE

Word count for abstract: 250
Word count for main text: 3922

Tables: 2
Figures: 3

Abstract

Aim: To analyze the effects of two resistance training (RT) programs that used the same relative loading but different repetition volume, using the velocity loss during the set as the independent variable: 15% (VL15) vs. 30% (VL30). **Methods:** Sixteen professional soccer players with RT experience (age 23.8 ± 3.5 years, body mass 75.5 ± 8.6 kg) were randomly assigned to two groups: VL15 ($n = 8$) or VL30 ($n = 8$) that followed a 6-week (18 sessions) velocity-based squat training program. Repetition velocity was monitored in all sessions. Assessments performed before (Pre) and after training (Post) included: estimated one-repetition maximum (1RM) and change in average mean propulsive velocity (AMPV) against absolute loads common to Pre and Post tests; countermovement jump (CMJ); 30-m sprint (T30); and Yo-yo intermittent recovery test (YYIRT). Null-hypothesis significance testing and magnitude-based inference statistical analyses were performed. **Results:** VL15 obtained greater gains in CMJ height than VL30 ($P < 0.05$), with no significant differences between groups for the remaining variables. VL15 showed a *likely/possibly* positive effect on 1RM (91/9/0%), AMPV (73/25/2%) and CMJ (87/12/1%), whereas VL30 showed *possibly/unclear* positive effects on 1RM (65/33/2%) and AMPV (46/36/18%) and *possibly negative* effects on CMJ (4/38/57%). The effects on T30 performance were *unclear/unlikely* for both groups, whereas both groups showed *most likely/likely* positive effects on YYIRT. **Conclusions:** A velocity-based RT program characterized by a low degree of fatigue (15% velocity loss in each set) is effective to induce improvements in neuromuscular performance in professional soccer players with previous RT experience.

Keywords: velocity-based resistance training, full squat, velocity specificity, athletic performance, training volume, strength training

Introduction

One of the main problems faced by strength and conditioning coaches is the issue of how to objectively quantify and monitor the actual training load undertaken by athletes in order to maximize performance.¹ There exist different methods to prescribe and monitor exercise intensity during resistance training (RT). Traditionally, the one-repetition maximum (1RM) has been considered the main reference to prescribe training directed towards developing strength and power abilities. However, during a RT program, athletes experience daily variations in neuromuscular performance and training readiness, and the actual 1RM values for a given subject and exercise may change from one training session to the next. Therefore, and since the current 1RM may not correspond with that measured on previous days or weeks, it cannot be ensured that the loads (%1RM) being used on each particular training session truly represent the intended ones. Another commonly used method is to prescribe loads from a test of maximum number of repetitions (nRM). This method implies that training sets are conducted to muscle failure, an approach which might not be optimal for some athletes.² Recently, velocity-based RT has been introduced. According to this novel approach, the training load for each session is set to match a given %1RM, which has its corresponding mean concentric velocity.¹ A pioneering study¹ analyzed the relationship between %1RM and mean propulsive velocity in the bench press. The extremely close relationship observed between %1RM and bar velocity ($R^2 = 0.98$) makes it possible to determine with considerable precision which %1RM is being used as soon as the first repetition of a set is performed with maximal voluntary velocity. Additional research has analyzed the load-velocity relationship in other exercises (prone bench pull, half-squat, squat, and leg press).³⁻⁶ All these studies have found strong relationships between loading magnitude and bar velocity, which allows the estimation of the 1RM value in each training session with a reasonable degree of accuracy.^{1,3-6} A very important practical application of

this methodology is the possibility of monitoring, in real-time, the actual load (%1RM) being used by measuring repetition velocity during training.^{1,3-6} Even more important is the fact that strength and conditioning coaches can observe the changes in strength that occur during the course of a training program, without the need to perform the often demanding, time-consuming and interfering 1RM assessments every few training sessions.¹ Interestingly, the predictive power of these equations ($R^2 = 0.96-0.98$) seems independent of the training background and the athletes' strength levels.^{1,4} Therefore, monitoring repetition velocity during training would allow to determine whether the proposed load (kg) truly represents the %1RM that was intended for each training session.

During RT in isoinertial conditions, and assuming every repetition is performed at maximal voluntary velocity, an unintentional decrease in force, velocity and hence power output is observed as fatigue develops and the number of repetitions approaches failure.⁷⁻⁸ It has been shown that monitoring repetition velocity is a practical and non-invasive way to estimate the acute metabolic stress, hormonal response, muscle damage, autonomic cardiovascular response and mechanical fatigue induced by RT.^{8,9,11} Thus, the repetition velocity loss experienced during each resistance set may serve as an objective indicator to monitor the actual degree of fatigue. A recent study¹⁰ has compared the effects of two squat training programs that only differed in the magnitude of repetition velocity loss allowed in each set: 20% vs. 40%. It was found that while a 40% velocity loss (which led to muscle failure in 56% of the training sets) could maximize the hypertrophic response, it also resulted in a fast-to-slow shift in muscle phenotype, whereas a velocity loss of 20% resulted in similar or even superior strength gains, especially in high-velocity actions such as the vertical jump. Furthermore, it has been observed that reductions in the ability to rapidly apply force up to 48 h following resistance exercise to failure can negatively interfere with other components of physical training.^{9,11}

In light of these considerations, instead of performing a fixed number of repetitions with a certain amount of weight, the velocity-based RT approach proposes to prescribe training in terms of two variables⁸: 1) first (usually fastest) repetition's mean velocity, which is intrinsically related to loading magnitude;^{1,3} and 2) the maximum percentage of velocity loss allowed in each set. Therefore, the aim of this study was to analyze the effects of two RT programs with the same loading magnitude but different volume, using the velocity loss during each set as the independent variable, defined as either 15% (VL15) or 30% (VL30).

Methods

Subjects

Twenty highly trained male soccer players (age 23.8 ± 3.4 yr, height 1.74 ± 0.07 m, body mass 75.5 ± 8.6 kg) from a professional soccer club volunteered to participate in this study. Typical in-season weekly training for this team included: specific soccer training (5 sessions), physical conditioning (3-4 sessions, of which 2 were strength training) and competitive play (1 game per week), totaling approximately 16 h per week on average. All subjects had RT experience and were accustomed to performing the full squat (SQ) exercise with correct technique. Subjects were randomly assigned to one of two groups, which differed only in the magnitude of repetition velocity loss allowed in each training set: 15% (VL15; $n = 10$) or 30% (VL30; $n = 10$). Only those players who complied with at least 85% of all training sessions were included in the statistical analyses. Due to injury or illness, four players missed too many training sessions or were absent from the post testing session. Thus, of the 20 initially enrolled players, sixteen players remained for statistical analyses (VL15, $n = 8$; VL30, $n = 8$). Once informed about the purpose, testing procedures and potential risks of the investigation, all subjects gave their voluntary written consent to participate. The present investigation was approved by the Research Ethics Committee of Pablo de Olavide

University, and was conducted in accordance with the Declaration of Helsinki. None of the subjects was taking drugs, medications or dietary supplements.

Experimental design

Subjects trained three times per week (48-72 h apart) over a 6-week period for a total of 18 sessions. A progressive RT program which comprised only the SQ exercise was used (**Table 1**). The two groups trained at the same relative loading magnitude (%1RM) in each session but differed in the maximum percent velocity loss reached in each exercise set (15% vs. 30%). As soon as the corresponding target velocity loss limit was exceeded, the set was terminated. Sessions were performed in a research laboratory under the direct supervision of the investigators, at the same time of day (± 1 h) for each subject and under controlled environmental conditions (20°C and 65% humidity). In addition, players performed their normal training routine for the duration of the present investigation. Both VL15 and VL30 groups were assessed on two occasions: before (Pre) and after (Post) the 6-week training intervention. Both Pre and Post testing took place in two sessions separated by 48 h. The first session comprised the sprinting, jumping and squat loading tests (performed in that order, interspersed with a 5 min pause, and described later in detail). The *Yo-Yo Intermittent Recovery Test* (YYIRT) was performed on the second session.

Testing procedures

Sprint and vertical jump tests

Vertical jump and sprint running ability were assessed as indicators of explosive force production and lower limb whole muscle dynamic performance. Players performed two maximal, 30 m indoor sprints, with a 3-min rest between sprints. A standing start with the lead-off foot placed 1 m behind the first timing gate was used. Sprint times were measured using photocells (Polifemo Radio Light, Microgate, Bolzano, Italy). The shortest time to

cover 30 m (T30) was recorded. Five maximal countermovement jumps (CMJ) with $\sim 90^\circ$ of knee flexion were performed, with 20 s rests between each jump. CMJ height was registered, the highest and lowest values were discarded, and the resulting average kept for analysis. Jump height was determined using an infrared timing system (Optojump, Microgate, Bolzano, Italy). The same standardized warm-up protocol which incorporated several sets of progressively faster 30 m running accelerations and some practice jumps was conducted at Pre and Post tests. Test-retest reliability measured by the coefficient of variation (CV) were 0.8% and 3.1% for T30 and CMJ, respectively. The intraclass correlation coefficients (ICCs) were 0.98 (95% confidence interval, CI: 0.95-0.99) for T30, and 0.98 (95% CI: 0.96-0.99) for CMJ.

Isoinertial squat loading test

A Smith machine (Multipower Fitness Line, Peroga, Murcia, Spain) was used for the isoinertial progressive loading test. The players performed the SQ from an upright position, descending at a controlled velocity (~ 0.50 - $0.70 \text{ m}\cdot\text{s}^{-1}$) until the top of the thighs were below the horizontal plane, then immediately reversed motion and ascended back to the upright position at maximal intended velocity. Initial load was set at 20 kg and was progressively increased in 10 kg increments until the attained mean propulsive velocity (MPV) was $\sim 1.00 \text{ m}\cdot\text{s}^{-1}$ (range: 0.96 - $1.04 \text{ m}\cdot\text{s}^{-1}$).¹² This resulted in a total of 6.4 ± 1.2 increasing loads performed by each subject. The subjects performed 3 repetitions with each load. The inter-set recovery time was 3 min. Warm-up consisted of 5 min of joint mobilization exercises, followed by two sets of six repetitions (3 min rest between sets) with a 10 kg load. An identical warm-up and progression of absolute loads for each subject was used in the Pre and Post tests. Strong verbal encouragement was provided to motivate participants to give a maximal effort. All velocity measures reported in this study correspond to the mean velocity

of the propulsive phase of each repetition; i.e. the mean propulsive velocity (MPV). The propulsive phase was defined as that fraction of the concentric phase during which barbell acceleration was greater than the acceleration due to gravity.¹³ Only the best repetition at each load, according to the criterion of fastest MPV, was considered for subsequent analysis. The following variables derived from this progressive loading test were used for analysis: a) estimated 1RM value, which was calculated from the MPV attained against the heaviest load of the test, as follows: $\%1RM = -2.185 \cdot MPV^2 - 61.53 \cdot MPV + 122.5$ ($R^2 = 0.96$; $SEE = 5.5\% \text{ 1RM}$),¹⁴ and b) average MPV attained against all absolute loads common to Pre and Post tests (AMPV). Since the change in movement velocity against the same absolute load is directly dependent on the force applied, an increase in repetition velocity is an indicator of strength improvement.¹ Thus, the AMPV value was used in an attempt to analyze the extent to which the two training interventions (VL15 vs. VL30) affected the SQ load-velocity relationship^{10,15}. A linear velocity transducer (T-Force System, Ergotech, Murcia, Spain) was used to measure bar velocity. Instantaneous velocity was sampled at 1,000 Hz and smoothed using a 4th order low-pass *Butterworth* filter with no phase shift and 10 Hz cut-off frequency. The system's software automatically calculated the relevant kinematics of every repetition, provided auditory and visual velocity feedback in real-time and stored data on disk for analysis. Mean relative error in the velocity measurements for this system was found to be <0.25%, whereas displacement was accurate to 0.5 mm. When simultaneously performing 30 repetitions with two devices (range: 0.3-2.3 m·s⁻¹ mean velocity), an ICC of 1.00 (95% CI: 1.00-1.00) and CV of 0.57% were obtained for MPV.⁸

Yo-Yo intermittent recovery test level 1

This test consists of 2 x 20 m shuttle runs at increasing speeds, with 10 s of active recovery between attempts. The test was carried out indoors, and the running pace was set

using a beep signal. The test ended when the subjects failed to reach the finish line at the beep signal on two consecutive occasions. The total distance covered was recorded as the final result of the test.¹⁶

Resistance training program

The descriptive characteristics of the RT program are presented in **Table 1**. Both VL15 and VL30 groups trained using only the SQ exercise, as previously described. Relative magnitude of training loads (%1RM) and number of sets and inter-set recovery periods (4 min) were kept identical for both groups in each training session. Relative loads were determined from the load-velocity relationship for the SQ since it has recently been shown that there is a very close relationship between %1RM and MPV.^{1,3,14} Thus, a target MPV to be attained in the first (usually the fastest) repetition of the first exercise set in each session was used as an estimation of %1RM, as follows: 1.13 m·s⁻¹ (~50% 1RM), 1.06 m·s⁻¹ (~55% 1RM), 0.98 m·s⁻¹ (~60% 1RM), 0.90 m·s⁻¹ (~65% 1RM), and 0.82 m·s⁻¹ (~70% 1RM); i.e. a velocity-based training was performed, instead of a traditional loading-based RT program.^{10,15,17} The absolute load (kg) was individually adjusted to match the velocity associated (± 0.03 m·s⁻¹) with the %1RM intended for each session. Loading magnitude progressively increased from 50 to 70% 1RM over the course of the study (**Table 1**). The groups differed in the degree of fatigue experienced during the exercise sets, which was objectively quantified by the magnitude of velocity loss attained in each set (15% vs. 30%) and, consequently, differed in the number of repetitions performed per set and the total number repetitions completed during the training program (**Table 1**). During training, subjects received immediate velocity feedback from the measurement system while being encouraged to perform each repetition at maximal intended velocity.

Statistical analyses

Values are reported as mean $\pm$ standard deviation (SD). Test-retest absolute reliability was assessed using the CV, whereas relative reliability was calculated using the ICC with a 95% CI, using the one-way random effects model. The normality of distribution of the variables in the Pre test and the homogeneity of variance across groups (VL15 vs. VL30) were verified using the *Shapiro-Wilk* test and *Levene's* test, respectively. Data were analyzed using a 2 x 2 factorial ANOVA using one between factor (VL15 vs. VL30) and one within factor (Pre vs. Post). Statistical significance was established at the $P \leq 0.05$ level. In addition to this null hypothesis testing, data were assessed for clinical significance using an approach based on the magnitudes of change.¹⁸⁻¹⁹ Effect sizes (ES) were calculated using *Hedge's g* on the pooled SD. Probabilities were also calculated to establish whether the true (unknown) differences were lower, similar or higher than the smallest worthwhile difference or change ($0.2 \times$ between-subject SD).²⁰ Quantitative chances of *better* or *worse* effects were assessed qualitatively as follows: $<1\%$, almost certainly not; 1-5%, very unlikely; 5-25%, unlikely; 25-75%, possible; 75-95%, likely; 95-99%, very likely; and $>99\%$, almost certain. If the chances of obtaining *beneficial/better* or *detrimental/worse* were both $>5\%$, the true difference was assessed as *unclear*.¹⁸⁻¹⁹ Inferential statistics based on the interpretation of magnitude of effects were calculated using a purpose-built spreadsheet for the analysis of controlled trials.²¹ The rest of the statistical analyses were performed using SPSS software version 18.0 (SPSS Inc., Chicago, IL).

Results

No significant differences between the two groups were found at Pre for any of the variables analyzed. Descriptive characteristics of the training actually performed by both groups are reported in **Table 1**. The repetitions performed in different velocity ranges by each group are shown in **Fig. 1**. Subjects in the VL15 group trained at a significantly faster mean

velocity than those in VL30 (0.91 ± 0.01 vs. 0.84 ± 0.02 m·s⁻¹, respectively; $P < 0.001$), whereas VL30 performed more repetitions ($P < 0.001$) than VL15 (414.6 ± 124.9 vs. 251.2 ± 55.4). Furthermore, VL30 completed more repetitions at slow velocities (0.4 - 0.9 m·s⁻¹) than VL15, whereas no differences between groups was found for the number of repetitions performed at high velocities (≥ 0.9 m·s⁻¹) (**Fig. 1**). The mean fastest repetition during each session, which indicates the %1RM of the load being lifted, did not differ between groups (0.98 ± 0.02 vs. 0.97 ± 0.02 m·s⁻¹, for VL30 and VL15, respectively). The actual mean velocity loss was $28.6 \pm 1.8\%$ for VL30 vs. $16.3 \pm 1.3\%$ for VL15. Mean repetition velocity attained in each set and training session for VL15 compared to VL30 is shown in **Fig. 2**.

Isoinertial strength assessments

Despite not finding ‘group’ x ‘time’ interactions for any of the isoinertial strength variables analyzed, practical worthwhile differences between the VL15 and VL30 training groups seemed evident as supported by the magnitude of the ES and qualitative outcomes (**Table 2**). VL15 showed a *likely/possibly* positive effect on 1RM strength and AMPV, respectively, whereas VL30 showed *possibly/unclear* positive effects on 1RM strength and AMPV, respectively. Furthermore, only VL15 showed significant improvements in 1RM strength ($P < 0.01$). **Fig. 3** shows the evolution of the estimated 1RM in each training session for both training groups, based on the relationship existing between MPV and %1RM in the SQ exercise.¹⁴

Vertical jump, sprint ability and endurance capacity

VL15 showed significantly greater gains in CMJ height than VL30 ($P < 0.05$), whereas no significant interaction was found for T30 and distance covered in the YYIRT. In addition, only the VL15 group improved CMJ height ($P < 0.05$), whereas both groups attained significant improvements in YYIRT ($P < 0.01$). The approach based on the

magnitudes of change showed a *likely* positive effect on CMJ height for VL15, whereas VL30 showed a *possibly negative* effect on CMJ performance (**Table 2**). The effects on T30 performance were *unclear/unlikely* for VL15 and VL30, respectively. The effects on YYIRT were *most likely/likely* positive effects for VL15 and VL30, respectively (**Table 2**).

Discussion

To our knowledge, this is the first study that has analyzed the effect of two velocity-based RT programs with the same loading magnitude (%1RM) but different training volume, using the velocity loss during the set as the independent variable (15% vs. 30%) in professional soccer players. An important aspect of this investigation was that movement velocity was measured and recorded for every repetition, using a linear velocity transducer. The strict control of the actual repetition velocities performed by the two experimental groups enabled us to isolate the effect of the variable of interest, in this case velocity loss, on the observed adaptations. The main finding of this study was that training with a velocity loss of 15% (VL15) in each set induced similar gains in squat performance (1RM strength as well as the velocity attained against all loads, from light to moderate) and endurance capacity (YYIRT), and greater gains in CMJ height, than training with a velocity loss of 30% (VL30). These results were observed despite the fact that the VL30 group performed significantly more repetitions than VL15 (415 vs. 251) during the training program. Even though both groups performed a similar number of repetitions at high velocities ($\geq 0.9 \text{ m}\cdot\text{s}^{-1}$), VL30 completed significantly more repetitions at slow velocities ($0.4\text{--}0.9 \text{ m}\cdot\text{s}^{-1}$) (**Fig. 1**). It could be argued that a lower degree of fatigue (velocity loss) would allow higher force application and hence faster repetition velocities during training. Therefore, setting a certain percent velocity loss threshold during RT seems a plausible way to avoid performing unnecessarily slow and fatiguing repetitions that may not contribute to the desired training effect.

Since the study conducted by Delorme,²² repetition to failure has been considered by many as a cornerstone of RT.²³⁻²⁵ However, recent evidence suggests that despite the high levels of discomfort and fatigue experienced in training to failure routines, non-failure training leads to similar or even greater gains in muscular strength.^{2,10,26-28} In this regard, it has recently been shown that a lower velocity loss during the set (20%) induces greater gains in performance, especially in high-velocity actions, when compared with RT characterized by high velocity loss (40%).¹⁰ In the squat exercise, a velocity loss of 40-50% in the set means that the set is conducted to, or very close to, muscle failure.^{8,10} In the present study, where muscle failure was not reached even in the VL30 group, the results seem to be in line with those findings¹⁰ since a velocity loss of 15% resulted in similar gains in performance than a velocity loss of 30%, and even greater gains in CMJ height. The present results also give support to previous studies that suggested the existence of an inverted U-shaped relationship between training volume and performance increase.²⁹⁻³¹ Therefore, once a certain amount of training volume (dose) is achieved, measured in this case by the velocity loss attained during the resistance exercise set, performing additional repetitions does not seem to elicit further strength gains and may even be detrimental for improving explosive strength.

The 1RM or nRM tests have been the most common methods to prescribe RT in soccer. However, this type of tests requires considerable effort from the subjects and may involve unnecessary risks and stress. In addition, the direct and precise measurement of 1RM can be difficult if movement velocity is not adequately monitored.¹ A novel velocity-based RT approach was therefore proposed in which the training load is adjusted based on movement velocity, due to the high correlation existing between %1RM and MPV ($R^2 = 0.96-0.98$).^{1,3-6} Previous studies have used this methodology with soccer players.^{12,32-34} However, in such studies the training load (kg) was established according to the velocity achieved against different loads during an initial squat loading test, and no further load

adjustments were performed during the training intervention. To our knowledge, the present study is the first to monitor the repetition velocity in each session during a RT program for soccer players. The estimated 1RM in each training session for every player (**Fig. 3**) shows that VL15 training resulted in an increased strength performance during almost all the training program, whereas the VL30 group showed similar performance to the Pre test values until session 7 and remained at a lower level of strength performance during most of the sessions when compared with VL15. This fact is very relevant in sports that require the maintenance of a high strength performance level throughout the season where competitions are held every weekend or even every 3-4 days. In addition, resistance exercise characterized by large reductions in repetition velocity, as it occurs in typical training to failure routines, requires longer recovery times,^{9,11} which is an important aspect to consider for most competitive athletes, since excessive fatigue resulting from RT could interfere with the development of other components of training.³⁵

Conclusions

Velocity-based RT characterized by a low degree of fatigue (15% velocity loss in each set) resulted in significant gains in squat strength and endurance performance, and even greater gains in CMJ height than a RT program that induced greater levels of fatigue (30% velocity loss), despite the VL30 group performing considerably more repetitions per set than the VL15 group (10.5 ± 1.9 vs. 6.0 ± 0.9 rep) against the same relative loads (%1RM). These findings emphasize the importance of finding an optimal dose during RT aimed at maximizing performance in competitive team sports and strongly suggest that often “less is more”. Indeed, squatting with a velocity loss of 30% during the set was found less effective and efficient than squatting with a velocity loss of 15% in professional soccer players. Taken

together, these results suggest that improvements in performance could be compromised when an excessive repetition volume is exceeded.

Practical applications

The results of the present study contribute to improve our knowledge about the process and methodology of load monitoring in resistance exercise. The magnitude of velocity loss attained during each training set may provide valid information about the optimal degree of fatigue necessary for maximizing performance. Thus, first repetition's mean velocity (which is intrinsically related to loading magnitude¹) and the percent velocity loss attained during the set,⁸ are two variables that should be monitored during a RT program. Velocity-based resistance training seems a novel, comprehensive and rational alternative to traditional RT.

References

1. González-Badillo JJ, Sánchez-Medina L. Movement velocity as a measure of loading intensity in resistance training. *Int J Sports Med.* 2010;31:347-352.
2. Davies T, Orr R, Halaki M, Hackett D. Effect of training leading to repetition failure on muscular strength: a systematic review and meta-Analysis. *Sports Med.* 2016; 46:487-502.
3. Sánchez-Medina L, González-Badillo JJ, Pérez CE, Pallarés JG. Velocity- and power-load relationships of the bench pull vs. bench press exercises. *Int J Sports Med.* 2014;35:209-216.
4. Loturco I, Pereira LA, Cal Abad CC, et al. Using the bar-velocity to predict the maximum dynamic strength in the half-squat exercise. *Int J Sports Physiol Perform.* 2015 [Epub ahead of print]
5. Conceicao F, Fernandes J, Lewis M, González-Badillo JJ, Jiménez-Reyes P. Movement velocity as a measure of exercise intensity in three lower limb exercises. *J Sports Sci.* 2015;34:1-8.
6. Bazuelo-Ruiz B, Padial P, García-Ramos A, Morales-Artacho AJ, Miranda MT, Feriche B. Predicting maximal dynamic strength from the load-velocity relationship in squat exercise. *J Strength Cond Res.* 2015;29:1999-2005.
7. Izquierdo M, González-Badillo JJ, Hakkinen K, et al. Effect of loading on unintentional lifting velocity declines during single sets of repetitions to failure during upper and lower extremity muscle actions. *Int J Sports Med.* 2006;27:718-724.
8. Sánchez-Medina L, González-Badillo JJ. Velocity loss as an indicator of neuromuscular fatigue during resistance training. *Med Sci Sports Exerc.* 2011;43:1725-1734.
9. González-Badillo JJ, Rodríguez-Rosell D, Sánchez-Medina L, et al. Short-term recovery following resistance exercise leading or not to failure. *Int J Sports Med.* 2016;37:295-304.
10. Pareja-Blanco F, Rodríguez-Rosell D, Sánchez-Medina L, et al. Effects of velocity loss during resistance training on athletic performance, strength gains and muscle adaptations. *Scand J Med Sci Sports.* 2016 [Epub ahead of print] doi: 10.1111/sms.12678
11. Pareja-Blanco F, Rodríguez-Rosell D, Sánchez-Medina L, et al. Acute and delayed response to resistance exercise leading or not leading to muscle failure. *Clin Physiol Funct Imaging.* 2016 [Epub ahead of print] doi: 10.1111/cpf.12348
12. González-Badillo JJ, Pareja-Blanco F, Rodríguez-Rosell D, Abad-Herencia JL, del Ojo-Lopez JJ, Sánchez-Medina L. Effects of velocity-based resistance training on young soccer players of different ages. *J Strength Cond Res.* 2015;29:1329-1338.
13. Sánchez-Medina L, Pérez CE, González-Badillo JJ. Importance of the propulsive phase in strength assessment. *Int J Sports Med.* 2010;31:123-129.

14. Sánchez-Medina L, García-Pallarés J, Pérez CE, Fernandes J, González-Badillo JJ. Estimation of relative load from mean velocity in the full squat exercise. In: *Cable NT, George K, eds. Book of Abstracts of the 16th Annual Congress of the European College of Sports Science*. Liverpool, UK: Liverpool John Moores University. 2011:669.
15. Pareja-Blanco F, Rodríguez-Rosell D, Sánchez-Medina L, Gorostiaga EM, González-Badillo JJ. Effect of movement velocity during resistance training on neuromuscular performance. *Int J Sports Med*. 2014;35:916-924.
16. Krstrup P, Mohr M, Amstrup T, et al. The yo-yo intermittent recovery test: physiological response, reliability, and validity. *Med Sci Sports Exerc*. 2003;35:697-705.
17. González-Badillo JJ, Rodríguez-Rosell D, Sánchez-Medina L, Gorostiaga EM, Pareja-Blanco F. Maximal intended velocity training induces greater gains in bench press performance than deliberately slower half-velocity training. *Eur J Sport Sci*. 2014;14:772-781.
18. Hopkins WG, Marshall SW, Batterham AM, Hanin J. Progressive statistics for studies in sports medicine and exercise science. *Med Sci Sports Exerc*. 2009;41:3-13.
19. Batterham AM, Hopkins WG. Making meaningful inferences about magnitudes. *Int J Sports Physiol Perform*. 2006;1:50-57.
20. Cohen J. Statistical power analysis for the behavioral sciences. *Hillsdale, MI: Lawrence Erlbaum*. 1988
21. Hopkins WG. Spreadsheets for analysis of controlled trials, with adjustment for a subject characteristic. *Sportscience*. 2006;10:46-50.
22. Delorme T. Restoration of muscle power by heavy-resistance exercises. *J Bone Joint Surg Am*. 1945;27:645-667.
23. Campos GE, Luecke TJ, Wendeln HK, et al. Muscular adaptations in response to three different resistance-training regimens: specificity of repetition maximum training zones. *Eur J Appl Physiol*. 2002;88:50-60.
24. Drinkwater EJ, Lawton TW, Lindsell RP, Pyne DB, Hunt PH, McKenna MJ. Training leading to repetition failure enhances bench press strength gains in elite junior athletes. *J Strength Cond Res*. 2005;19:382-388.
25. Ahtiainen JP, Pakarinen A, Kraemer WJ, Hakkinen K. Acute hormonal and neuromuscular responses and recovery to forced vs maximum repetitions multiple resistance exercises. *Int J Sports Med*. 2003;24:410-418.
26. Folland JP, Irish CS, Roberts JC, Tarr JE, Jones DA. Fatigue is not a necessary stimulus for strength gains during resistance training. *Br J Sports Med*. 2002;36:370-374.

27. Izquierdo-Gabarren M, González de Txabarri Expósito R, García-Pallarés J, Sánchez-Medina L, de Villarreal ES, Izquierdo M. Concurrent endurance and strength training not to failure optimizes performance gains. *Med Sci Sports Exerc.* 2010;42:1191-1199.
28. Izquierdo M, Ibáñez J, González-Badillo JJ, et al. Differential effects of strength training leading to failure versus not to failure on hormonal responses, strength, and muscle power gains. *J Appl Physiol.* 2006;100:1647-1656.
29. Kuipers H. How much is too much? Performance aspects of overtraining. *Res Q Exerc Sport.* 1996;67:S65-69.
30. González-Badillo JJ, Gorostiaga EM, Arellano R, Izquierdo M. Moderate resistance training volume produces more favorable strength gains than high or low volumes during a short-term training cycle. *J Strength Cond Res.* 2005;19:689-697.
31. González-Badillo JJ, Izquierdo M, Gorostiaga EM. Moderate volume of high relative training intensity produces greater strength gains compared with low and high volumes in competitive weightlifters. *J Strength Cond Res.* 2006;20:73-81.
32. Franco-Marquez F, Rodríguez-Rosell D, González-Suárez JM, et al. Effects of combined resistance training and plyometrics on physical performance in young soccer players. *Int J Sports Med.* 2015;36:906-914.
33. López-Segovia M, Palao Andrés JM, González-Badillo JJ. Effect of 4 months of training on aerobic power, strength, and acceleration in two under-19 soccer teams. *J Strength Cond Res.* 2010;24:2705-2714.
34. Rodríguez-Rosell D, Franco-Márquez F, Pareja-Blanco F, et al. Effects of 6-weeks resistance training combined with plyometric and speed exercises on physical performance of pre-peak height velocity soccer players. *Int J Sports Physiol Perform.* 2015;11:240-246.
35. Draganidis D, Chatzinikolaou A, Jamurtas AZ, et al. The time-frame of acute resistance exercise effects on football skill performance: the impact of exercise intensity. *J Sports Sci.* 2013;31:714-722.

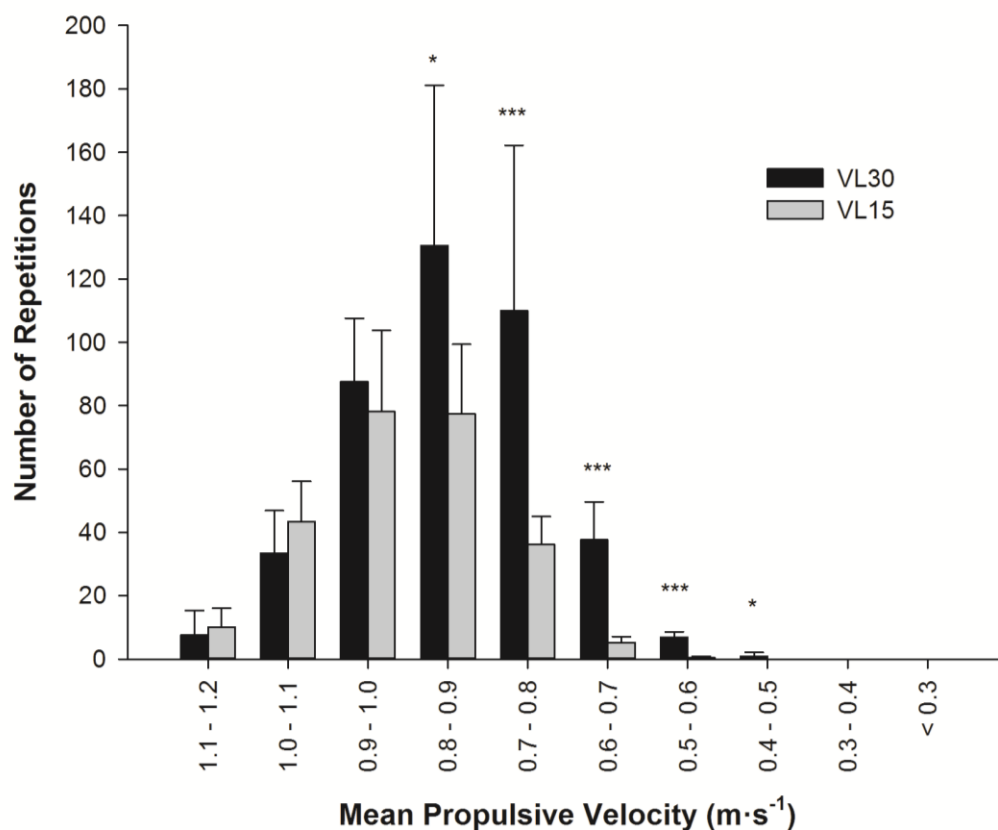


Figure 1—Number of repetitions in the squat exercise performed in each velocity range by both training groups. Data are mean $\pm$ SD. Statistically significant differences between groups: * $P < 0.05$, *** $P < 0.001$. VL15: group that trained with a mean velocity loss of 15% in each set ($n = 8$); VL30: group that trained with a mean velocity loss of 30% in each set ($n = 8$). Warm-up repetitions are excluded.

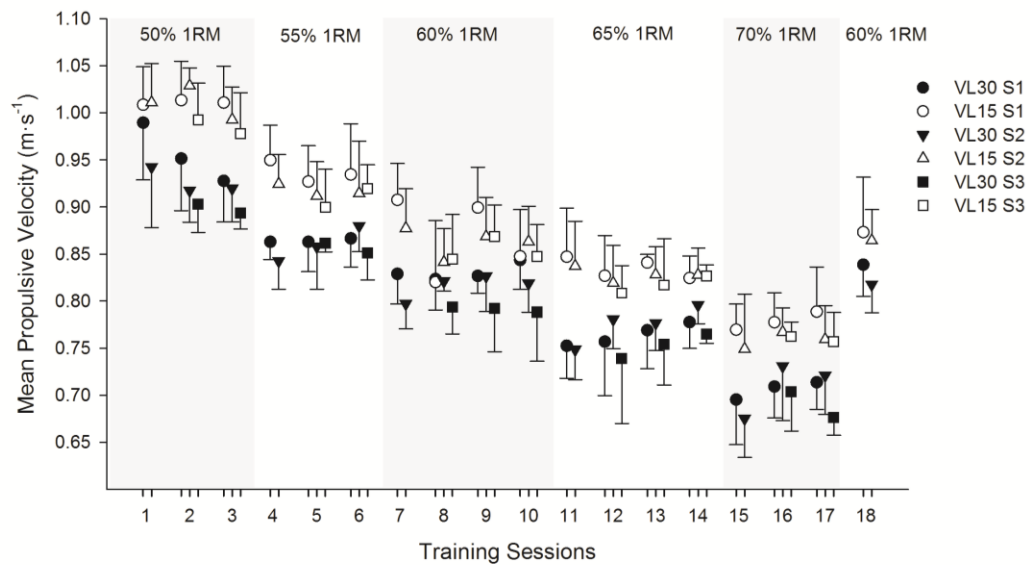


Figure 2– Mean repetition velocity attained in each set and training session for VL15 compared to VL30. Data are mean $\pm$ SD. VL15: group that trained with a mean velocity loss of 15% in each set ($n = 8$); VL30: group that trained with a mean velocity loss of 30% in each set ($n = 8$). Warm-up repetitions are excluded.

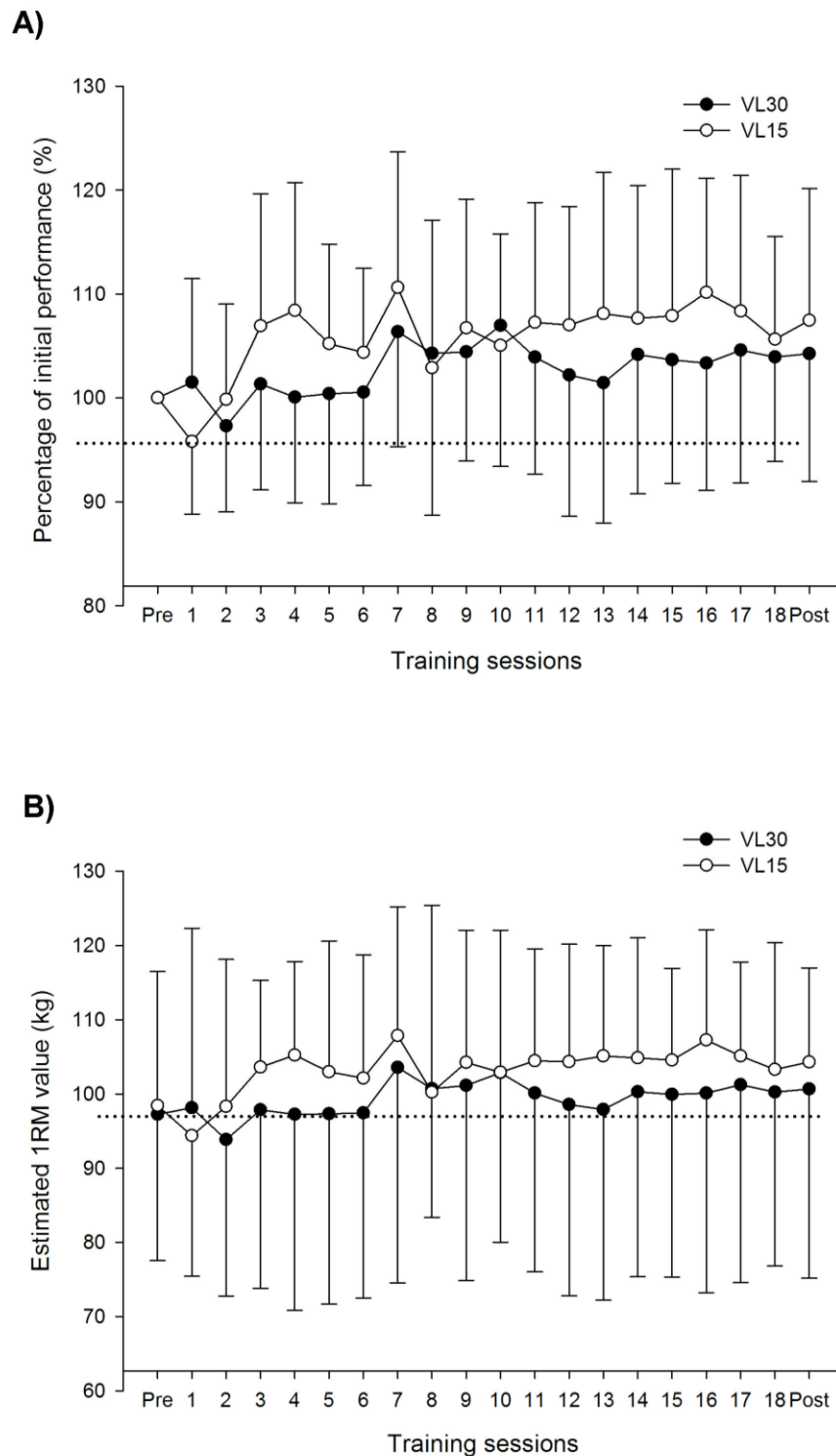


Figure 3—Evolution of the estimated 1RM strength in the squat exercise in each training session expressed as: (A) Percentage of the initial Pre-training level; and (B) absolute load (kg). Data are mean $\pm$ SD. VL15: group that trained with a mean velocity loss of 15% in each set ($n = 8$); VL30: group that trained with a mean velocity loss of 30% in each set ($n = 8$).

Table 1. Descriptive characteristics of the 6-week velocity-based squat training program performed by both experimental groups.

<i>Scheduled</i>	Session 1	Session 2	Session 3	Session 4	Session 5	Session 6	Session 7	Session 8	Session 9
Sets x VL (%)									
VL15	2 x 15%	3 x 15%	3 x 15%	2 x 15%	3 x 15%	3 x 15%	2 x 15%	3 x 15%	3 x 15%
VL30	2 x 30%	3 x 30%	3 x 30%	2 x 30%	3 x 30%	3 x 30%	2 x 30%	3 x 30%	3 x 30%
Target MPV ($\text{m}\cdot\text{s}^{-1}$)	1.13 (~50% 1RM)	1.13 (~50% 1RM)	1.13 (~50% 1RM)	1.06 (~55% 1RM)	1.06 (~55% 1RM)	1.06 (~55% 1RM)	0.98 (~60% 1RM)	0.98 (~60% 1RM)	0.98 (~60% 1RM)
<i>Scheduled</i>	Session 10	Session 11	Session 12	Session 13	Session 14	Session 15	Session 16	Session 17	Session 18
Sets x VL (%)									
VL15	3 x 15%	2 x 15%	3 x 15%	3 x 15%	3 x 15%	2 x 15%	3 x 15%	3 x 15%	2 x 15%
VL30	3 x 30%	2 x 30%	3 x 30%	3 x 30%	3 x 30%	2 x 30%	3 x 30%	3 x 30%	2 x 30%
Target MPV ($\text{m}\cdot\text{s}^{-1}$)	0.98 (~60% 1RM)	0.90 (~65% 1RM)	0.90 (~65% 1RM)	0.90 (~65% 1RM)	0.90 (~65% 1RM)	0.82 (~70% 1RM)	0.82 (~70% 1RM)	0.82 (~70% 1RM)	0.98 (~60% 1RM)
<i>Actually Performed</i>	Fastest MPV ($\text{m}\cdot\text{s}^{-1}$)	MPV all reps ($\text{m}\cdot\text{s}^{-1}$)	Total rep	Rep per set	Rep per set with 50% 1RM	Rep per set with 55% 1RM	Rep per set with 60% 1RM	Rep per set with 65% 1RM	Rep per set with 70% 1RM
VL15	0.97 ± 0.02	0.91 ± 0.01	251.2 ± 55.4	6.0 ± 0.9	10.9 ± 2.0	6.1 ± 1.4	5.0 ± 1.1	4.8 ± 1.6	4.1 ± 1.1
VL30	0.98 ± 0.02	0.84 ± 0.02***	414.6 ± 124.9***	10.5 ± 1.9***	14.7 ± 2.3**	11.9 ± 2.6***	9.5 ± 1.9***	9.1 ± 3.1**	7.2 ± 2.1**

Data are mean ± SD. Only one exercise (full squat) was used in training.

VL15: Group that trained with a mean velocity loss of 15% in each set (n = 8), VL30: Group that trained with a mean velocity loss of 30% in each set (n = 8)

MPV: Mean Propulsive Velocity

VL: Velocity loss in the set calculated as a percent loss in MPV from the fastest (usually first) to the slowest (last one) repetition of each set

Target MPV: MPV scheduled for the first repetition of the first set in each session, which corresponds with the loading magnitude (%1RM) intended for that session

Fastest MPV: Average of the fastest repetition measured in each session (this value is an indicator of the average loading magnitude, %1RM, achieved during the training program)

MPV all reps: Average MPV attained during the entire training program

Total rep: Total number of repetitions performed during the training program

Rep per set: average number of repetitions performed in each set

Rep per set with a given %1RM: average number of repetitions performed in each set with each of the loads used (50-70 %1RM).

Significant differences between VL15 and VL30 groups in mean overall values: ** P < 0.01; *** P < 0.001

Table 2. Changes in selected neuromuscular performance variables from Pre- to Post-training.

	Pre	Post	ES (90% CI)	Percent changes of better/trivial/worse effect	
1RM-VL15 (kg)	101.3 ± 18.8	110.3 ± 14.3**	0.43 (0.14 to 0.71)	91/9/0	Likely
1RM-VL30 (kg)	100.2 ± 20.3	106.5 ± 28.5	0.28 (-0.09 to 0.64)	65/33/2	Possibly
AMPV-VL15 (m·s ⁻¹)	1.19 ± 0.12	1.23 ± 0.09	0.35 (-0.09 to 0.79)	73/25/2	Possibly
AMPV-VL30 (m·s ⁻¹)	1.16 ± 0.11	1.18 ± 0.13	0.16 (-0.55 to 0.87)	46/36/18	Unclear
CMJ-VL15 (cm)	33.7 ± 3.6	35.5 ± 5.1*†	0.45 (0.06 to 0.85)	87/12/1	Likely
CMJ-VL30 (cm)	34.4 ± 3.5	33.5 ± 3.1	-0.24 (-0.66 to 0.18)	4/38/57	Possibly Negative
T30-VL15 (s)	4.32 ± 0.19	4.30 ± 0.20	0.10 (-0.14 to 0.35)	24/74/3	Unlikely
T30-VL30 (s)	4.28 ± 0.14	4.27 ± 0.10	0.06 (-0.27 to 0.39)	21/70/9	Unclear
YYIRT-VL15 (m)	1390 ± 417	1862 ± 639**	1.01 (0.63 to 1.39)	100/0/0	Most Likely
YYIRT-VL30 (m)	1611 ± 422	2043 ± 842**	0.97 (0.13 to 1.82)	94/4/2	Likely

Data are mean ± SD; ES = within-group Effect Size; CI = Confidence Interval

VL15: group that trained with a mean repetition velocity loss of 15% in each set (n = 8)

VL30: group that trained with a mean repetition velocity loss of 30% in each set (n = 8)

1RM: estimated one-repetition maximum squat strength

AMPV: average MPV attained against absolute loads common to Pre- and Post-tests in the squat progressive loading test

CMJ: countermovement jump height

T30: 30 m sprint running time

YYIRT: Yo-yo intermittent recovery test level 1

Intra-group significant differences from Pre- to Post-training: * P < 0.05, ** P < 0.01

Significant group x time interaction: † P < 0.05